



FIRST SEMESTER, 2026–2027
Course Handout Part II

Date: 8th August 2026

In addition to Part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course No.	: MAC F213
Course Title	: Discrete Mathematics
Instructor-in-Charge	: DIVYUM SHARMA
Instructors	:

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1. Course Description:

Set theory: Sets, relations and functions, equivalence relations, partially ordered sets, countability of sets, lattices and Boolean algebras;

Logic and methods of proof: Propositional logic, language of propositional logic, predicate logic and logical inference with quantifiers, introduction to different standard proof techniques;

Combinatorics: Counting techniques, pigeonhole principle, inclusion-exclusion principle, recurrence relations, generating function, partitions;

Graph theory: Graphs and digraphs, special types of graphs, trees, isomorphism, connectedness, Euler and Hamilton graphs.

2. Scope & Objective of the Course:

Mathematics, described as a language of science, has acquired its unique position due to its precision and rigour. This makes essential the development of the sense for mathematical rigor as well as the habit of mathematical thought process. The course will achieve this by introducing students to propositional and predicate logic. As an important follow-up, various methods of proof will be discussed. Several mathematical structures like relations and orderings are studied due to their importance, not only in mathematics but also applied subjects like computer science. The course is also useful for the preparation for the study of computational concepts, techniques, and skills necessary to comprehend the structure of problems encountered in the design and analysis of algorithms.

3. Course learning outcomes

Upon successful completion of this course, students should be able to:

- CLO1.** understand the basic process of mathematical reasoning
- CLO2.** identify valid arguments and fallacies;
- CLO3.** understand the structure of proofs of theorems in Mathematics;

- CLO4.** write valid mathematical proofs;
- CLO5.** use powerful techniques for handling computing problems in Mathematics and Computation;
- CLO6.** understand important structures having wide applications to sciences, computer programming and various classification problems.;
- CLO7.** to create new relations with useful properties from a given relation;
- CLO8.** to learn basic concepts of Graph theory.

4. Textbooks:

TB1. *Discrete Mathematics & its Applications*, K. H. Rosen, 8th Edition, McGraw Hill Education (India) Pvt. Ltd., 2011, Reprint 2021.

References:

Ref1. *Discrete Mathematics for Computer Scientists & Mathematicians*, Mott, Kandel & Baker, 2nd Edition, PHI, 2010.

Ref2. *Discrete Mathematics with Graph Theory*, E.G. Goodaire & M.M. Parmenter, 2nd Edition, PHI, 2002.

5. Course Plan:

Module	Description	Lecture No.	Section No.	Learning Outcomes
1. Basics of reasoning and arguments	Introduction & Overview	1	1	To understand the basic process of mathematical reasoning, valid arguments and fallacies.
	Propositional Logic	2-3	Section 1.1 (1.1.1-1.1.4)	
	Propositional Equivalences	4	Section 1.3 (1.3.1-1.3.4)	
	Predicates & quantifiers	5	Section 1.4	
	Nested quantifiers	6	Section 1.5	
	Rules of inference	7	Section 1.6 (1.6.1-1.6.4, 1.6.7-1.6.9)	
2. Methods of proofs in Mathematics	Methods of proof	8-9	Sections 1.7, 1.8	To understand the structure of proofs of theorems in
	Mathematical induction, strong form of	10-11	Sections 5.1, 5.2	

	mathematical induction			Mathematics
3. Introduction to combinatorics	Pigeonhole principle	12-13	Section 6.2	To develop powerful techniques for handling computing problems in Mathematics & Computation
	Generalized Combinations and Permutations	14-16	Sections 6.3, 6.5, 6.6	
	Binomial Identities	Self-study	Section 6.4	
	Recurrence relations	17	Section 7.1	
	Solving linear recurrence relations	18-19	Section 7.2	
	Generating functions	20	Section 7.4	
	Solving linear recurrence relations using generating functions	21-22	Section 7.4.5	
	Inclusion – Exclusion principle	23-24	Section 7.5	
4. Relations on a Set, their representations and important Properties	Relations, equivalence relations, Operations on Relations, paths and closures, Digraphs	25-28	Sections 8.1 – 8.5	To understand important structures having wide applications to sciences, computer programming and various classification problems. Learn to create nice relations from a given relation.
	Partially ordered set, Hasse diagrams, Lattice Theory	29-30	Section 8.6	
5. Introduction to Graph Theory	Graphs and subgraphs	31-32	Sections 9.1, 9.2	Studying basic concepts of Graph theory
	Graph isomorphism	33-34	Section 9.3	
	Connected graphs and trees	35-37	Section 9.4	
	Euler & Hamiltonian paths	38-40	Section 9.5	

6. Evaluation Scheme:

Component	Wt (%)	Duration	Date & Time	Comments
Quizzes	20	30 mins. each		Closed book; 3 announced quizzes; best 2 out of 3 quiz scores will be consid- ered
Mid-Semester	35	90 mins.	05/10 AN2 or, as announced by AUGSD	Closed Book
Comprehensive Examination	45	180 mins.	03/12 AN or, as announced by AUGSD	Open book/Closed book

7. NC Criterion:

Any student scoring less than 20 marks out of 200 will be given NC grade. Further, if a student is absent in any evaluative component (that is, either in mid-semester or comprehensive exam or in all the quizzes), then he/she will be given NC grade.

8. Attendance Policy:

Each student is expected to attend all classes and appear in all evaluative components.

9. Chamber Consultation Hour:

To be announced by the respective instructor. The chamber consultation hours of the instructors will be up-loaded on the NALANDA website.

10. Make-up Policy:

Make-up for mid-semester and comprehensive exams will be given only for very genuine cases, and prior permission must be obtained from IC. No makeup will be given for quizzes under any circumstances whatsoever.

11. Notices:

The notices concerning this course will be displayed on the NALANDA website.

Instructor-in-Charge
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